ISEN/DAEN 427 **Mock Exam #1** **9/23/25**

Note: This will be a closed-book exam. You may use one two-sided, letter-sized sheet of notes and a scientific calculator. Smartphones, tablets, smartwatches, laptops, and notebooks will not be allowed. However in the mock exam you are free to use any resources you like.

1. Consider the following simplified version of the broken telephone game. A first player secretly communicates a message $m_1 \in \{0, 1, 2\}$ to a second player with probabilities

$$P(m_1 = 0) = P(m_1 = 1) = P(m_1 = 2) = \frac{1}{3}$$

The second player then secretly communicates a message $m_2 = m_1 + n$ to a third (final) player

$$P(n = 0) = \frac{2}{3} \quad P(n = +1) = \frac{1}{3}$$

   a. What is the optimal Bayes classification of the message received by the last player?
   Answer:

   – Player 1 chooses a message $m_1 \in \{0, 1, 2\}$ with equal probability

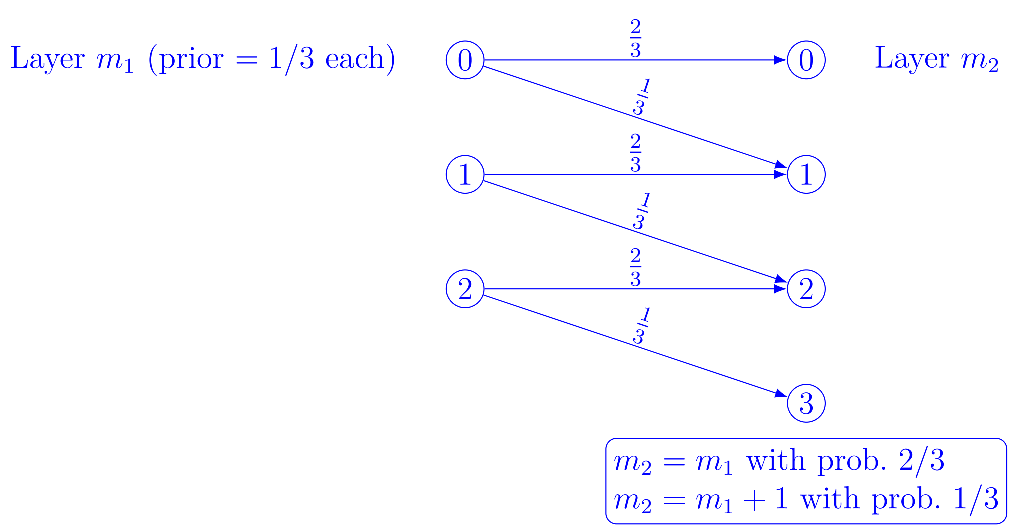
$$P(m_1 = 0) = P(m_1 = 1) = P(m_1 = 2) = \tfrac{1}{3}$$

   – Player 2 receives m_1 and then sends a noisy version m_2 to Player 3:

$$P(m_2 = m_1 \mid m_1) = \tfrac{2}{3}, \qquad P(m_2 = m_1 + 1 \mid m_1) = \tfrac{1}{3}$$

   Thus $m_2 \in \{0, 1, 2, 3\}$

   – Player 3 observes m_2 and must infer m_1



Step 1: Posterior distribution

$$P(m_1 \mid m_2) = \frac{P(m_2 \mid m_1)P(m_1)}{\sum_{a=0}^{2} P(m_2 \mid m_1 = a)P(m_1 = a)}.$$

Since $P(m_1 = a) = \frac{1}{3}$ for all a, the denominator acts as a normalizing constant.

Step 2: Case analysis

Case $m_2 = 0$:

$$\text{Denominator} = \tfrac{2}{3} \cdot \tfrac{1}{3} = \tfrac{2}{9}.$$

$$P(m_1 = 0 \mid m_2 = 0) = \frac{\frac{2}{3} \cdot \frac{1}{3}}{\frac{2}{9}} = 1$$

Hence only $m_1 = 0$ is possible.

Case $m_2 = 1$:

$$\text{Denominator} = \tfrac{2}{3} \cdot \tfrac{1}{3} + \tfrac{1}{3} \cdot \tfrac{1}{3} = \tfrac{2}{9} + \tfrac{1}{9} = \tfrac{1}{3}$$

$$P(m_1 = 1 \mid m_2 = 1) = \frac{\frac{2}{3} \cdot \frac{1}{3}}{\frac{1}{3}} = \frac{2}{3} \qquad P(m_1 = 0 \mid m_2 = 1) = \frac{\frac{1}{3} \cdot \frac{1}{3}}{\frac{1}{3}} = \frac{1}{3}$$

Case $m_2 = 2$:

$$\text{Denominator} = \tfrac{2}{3} \cdot \tfrac{1}{3} + \tfrac{1}{3} \cdot \tfrac{1}{3} = \tfrac{1}{3}.$$

$$P(m_1 = 2 \mid m_2 = 2) = \tfrac{2}{3} \qquad P(m_1 = 1 \mid m_2 = 2) = \tfrac{1}{3}$$

Case $m_2 = 3$:

$$\text{Denominator} = \tfrac{1}{3} \cdot \tfrac{1}{3} = \tfrac{1}{9}$$

$$P(m_1 = 2 \mid m_2 = 3) = 1$$

Step 3: Posterior table

Summarizing, the posterior distribution is

	$m_2 = 0$	$m_2 = 1$	$m_2 = 2$	$m_2 = 3$
$P(m_1 = 0 \mid m_2)$	1	$\frac{1}{3}$	0	0
$P(m_1 = 1 \mid m_2)$	0	$\frac{2}{3}$	$\frac{1}{3}$	0
$P(m_1 = 2 \mid m_2)$	0	0	$\frac{2}{3}$	1

Step 4: Bayes classifier

The Bayes estimate is to choose the m_1 with maximum posterior probability (MAP):

$$\hat{m}_1(m_2) = \begin{cases} 0, & m_2 = 0, \\ 1, & m_2 = 1, \\ 2, & m_2 = 2, \\ 2, & m_2 = 3. \end{cases}$$

Interpretation: the final player should generally trust the received value m_2. The only exception is when $m_2 = 3$, which could only have come from $m_1 = 2$.

b. What is the expected error probability?

Answer: From the posterior table:

$$P(\text{error} \mid m_2 = 0) = 0, \qquad P(\text{error} \mid m_2 = 1) = \tfrac{1}{3},$$
$$P(\text{error} \mid m_2 = 2) = \tfrac{1}{3}, \qquad P(\text{error} \mid m_2 = 3) = 0.$$

We also need the marginal distribution of m_2 (the denominators above):

$$P(m_2) = \sum_{a=0}^{2} P(m_2 \mid m_1 = a) P(m_1 = a), \quad P(m_1 = a) = \tfrac{1}{3}.$$

$$P(m_2 = 0) = \tfrac{2}{9}, \quad P(m_2 = 1) = \tfrac{1}{3}, \quad P(m_2 = 2) = \tfrac{1}{3}, \quad P(m_2 = 3) = \tfrac{1}{9}.$$

Therefore, the total (expected) error probability is

$$\begin{aligned} P(\text{error}) &= \sum_{m_2} P(m_2)\, P(\text{error} \mid m_2) \\ &= \left(\tfrac{2}{9} \cdot 0\right) + \left(\tfrac{1}{3} \cdot \tfrac{1}{3}\right) + \left(\tfrac{1}{3} \cdot \tfrac{1}{3}\right) + \left(\tfrac{1}{9} \cdot 0\right) \\ &= \tfrac{1}{9} + \tfrac{1}{9} = \boxed{\tfrac{2}{9}} \approx 0.222\ldots \end{aligned}$$

Or simply:

$$P(m_1 = 0 | m_2 = 1) P(m_2 = 1) + P(m_1 = 1 | m_2 = 2) P(m_2 = 2) = \frac{1}{3}\frac{1}{3} + \frac{1}{3}\frac{1}{3} = \frac{2}{9}$$

2. Many air travelers have waited expectantly in an airport baggage claim area, watching for their bags to drop down the chute into the circulating luggage carousel (figure below). This situation presents opportunities for probabilistic reasoning as well as perceptual inference. Let us suppose that you are engaged in this ritual of modern-day air travel along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage. A recording piped through the speakers reminds you that *"Many bags look alike. Please check your bag carefully before exiting the terminal."* Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case used by 5 percent of all travelers. Given your distance from the luggage chute, you can discern only the shape, size and color of the bags that emerge, not any detailed markings on the bags. Now let us suppose that the first bag from your flight to enter the luggage carousel indeed has the same shape, size, and color as your bag.

a. Is it yours? Explain.

Answer: There are 100 passengers, each with one bag. Your bag is a medium black rectangular suitcase, a style used by 5% of travelers. This means that about 5 bags on the flight (including yours) have the same appearance.

We want the probability that the first bag you see on the carousel, which matches your bag's size, shape, and color, is actually yours $P(\text{yours} \mid \text{looks like yours})$.

By Bayes' rule,

$$P(\text{Yours} \mid \text{looks like yours}) = \frac{P(\text{looks like yours} \mid \text{yours}) \cdot P(\text{yours})}{P(\text{looks like yours})}$$

- $P(\text{yours}) = \frac{1}{100}$ (only one of the 100 bags is yours),
- $P(\text{looks like yours} \mid \text{yours}) = 1$,
- $P(\text{looks like yours}) = 0.05$ (5% of all bags look like yours).

Thus, $P(\text{yours} \mid \text{looks like yours}) = \frac{1 \cdot \frac{1}{100}}{0.05} = \frac{0.01}{0.05} = 0.20$.

So, the probability that the first look-alike bag is yours is only 20%. $\boxed{\textit{WRONG!}}$

$$\begin{aligned} H_1 &: \text{ the bag is mine,} & P(H_1) &= 0.01 \\ H_2 &: \text{ the bag is not mine,} & P(H_2) &= 0.99 \end{aligned}$$

$$P(\text{obs} \mid H_1) = 1, \qquad P(\text{obs} \mid H_2) = 0.05$$

$$\begin{aligned} P(H_1 \mid \text{obs}) &= \frac{P(\text{obs} \mid H_1)P(H_1)}{P(\text{obs} \mid H_1)P(H_1) + P(\text{obs} \mid H_2)P(H_2)} \\ &= \frac{1 \cdot 0.01}{1 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.01}{0.0595} \approx 0.168 \end{aligned}$$

b. How the probability changes as more look-alikes appear?

Answer: Initially there are 5 look-alike bags (including yours) among the 100 total bags. The bags come out in random order.

Case 1: No look-alikes seen yet The first look-alike to appear is equally likely to be any of the 5. Therefore, $P(\text{first look-alike is yours}) = \frac{1}{5} = 20\%$.

Case 2: After t look-alikes have already appeared and none were yours There are $5-t$ look-alikes left, one of which is yours. The next look-alike to appear is equally likely to be any of these. Thus, $P(\text{next look-alike is yours} \mid t \text{ already seen}) = \frac{1}{5-t}$.

$$t=0 : \frac{1}{5} = 20\%, \quad t=1 : \frac{1}{4} = 25\%, \quad t=2 : \frac{1}{3} \approx 33.3\%, \quad t=3 : \frac{1}{2} = 50\%, \quad t=4 : 1 = 100\%$$

At first, there is only a 20% chance the first look-alike is yours. But with each non-matching look-alike you eliminate, the probability for the next one increases, eventually reaching 100% when only your bag remains. $\boxed{WRONG!}$

$$P(H_1 \mid \text{obs}) = \frac{1 \cdot \frac{1}{\text{total of bags remaining}}}{1 \cdot \frac{1}{\text{total of bags remaining}} + 0.05 \cdot \frac{\text{total of bags remaining}-1}{\text{Total of bags remaining}}}$$

if after 85 bags have come out and none was yours, 15 still remains

$$= \frac{1 \cdot \frac{1}{15}}{1 \cdot \frac{1}{15} + 0.05 \cdot \frac{14}{15}}$$

$$= \frac{1}{1 + 0.05 \cdot 14} = \frac{1}{1.7} \approx 0.588$$